

Nanoscale gold illustrates the unique properties that occur at the nanoscale. Nanoscale gold particles are not the yellow color with which we're familiar; they appear red or purple. At the nanoscale, the motion of the gold's electrons is confined. Because this movement is restricted, gold nanoparticles react differently to light compared to larger-scale gold particles. Their size and optical properties make them ideal candidates for practical use in tumour treatment. These nanoscale gold particles selectively accumulate in tumors, where they can enable both, precise imaging and targeted laser destruction of the tumor by means that avoid harming healthy cells.

A fascinating and powerful result of the quantum effects of the nanoscale is the concept of “tunability” of properties. That is, by changing the size of the particle, a scientist can literally fine-tune a material's property of interest (e.g., changing fluorescence colour; in turn, the fluorescence color of a particle can be used to identify the particle, and various materials can be “labelled” with fluorescent markers for various purposes). Another potent quantum effect of the nanoscale is known as “tunneling”. The scanning tunneling microscope and flash memory is based on this technology

## Biology

Over millennia, nature has perfected the art of biology at the nanoscale. A strand of DNA, one of the building blocks of human life, is only about two nanometres in diameter.

Drawing on the natural nanoscale of biology, many medical researchers are working on designing tools, treatments and therapies that are more precise and personalised than conventional ones – and that can be applied earlier in the course of a disease thus leading to fewer adverse side-effects. Some scientists are looking at ways to use nanoscale biological principles of molecular self-assembly, self-organisation and quantum mechanics to create novel computing platforms.

## Surface areas

Nanoscale materials have far larger surface areas than similar masses of larger-scale materials. We'll illustrate this with a simple thought experiment:

A solid cube of a material 1 cm on a side has 6 cms<sup>2</sup> of surface area, about equal to one side of half a stick of gum.

But if that volume of 1 cm<sup>3</sup> were filled with cubes 1 mm on a side, that would be 1,000 mm-sized cubes (10 x 10 x 10), each one of which has a surface area of 6 mms<sup>2</sup>, for a total surface area of 60 cms<sup>2</sup> – about the same as two thirds